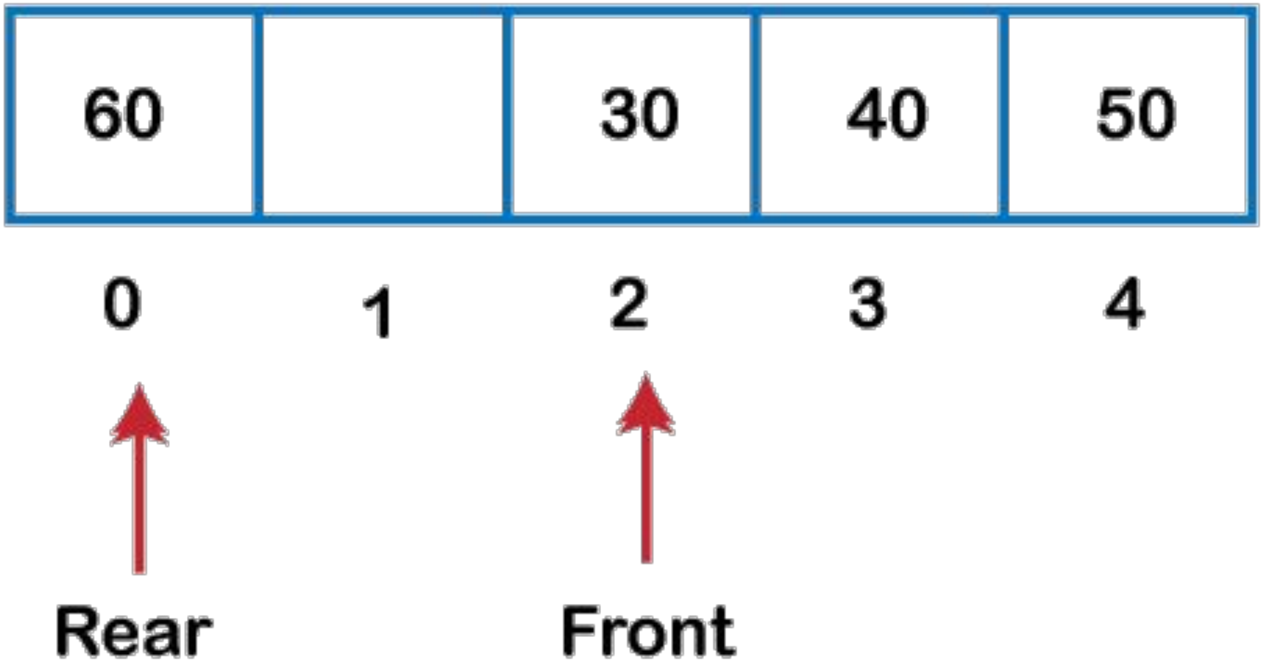
Data Structures

Lecture 11
Queue with Circular Array

Prantik Paul [PNP]
Lecturer
Department of Computer Science and Engineering
BRAC University

Circular Array



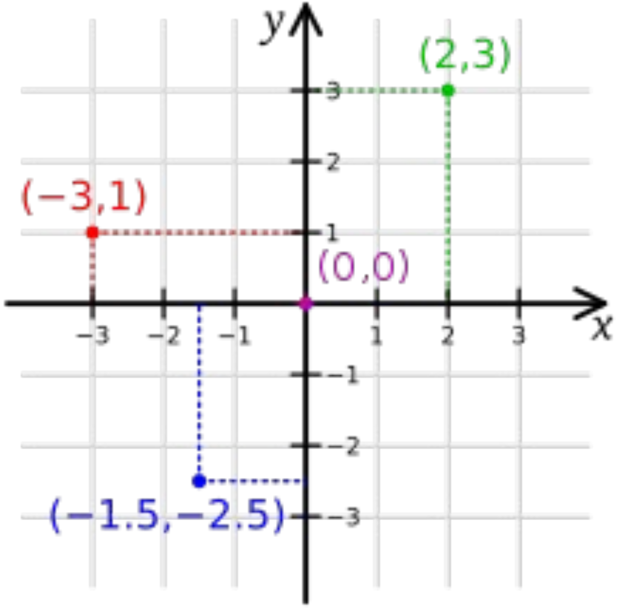
Circular Array (Indexing)

a = buffer[4 % 5]

a = buffer[(4 + 2) % 5]

a = buffer[(0 - 2) %5]

Circular Array (Indexing -2D)



element = grid[(i % 2M]][j % 2N]

Circular Array (Structure)

```
  0     1     2     3     4       5     6    7
----------------------------------------------   capacity = 8
| 5 |   7 |    9 |   15 | -4 |    17 |    |    |   size     = 6
----------------------------------------------
  ^-- start                               ^-- nextAvailPos

  0     1     2     3     4       5     6    7
----------------------------------------------   capacity = 8
|     | 5   |    7 |    9   | 15 | -4   | 17|    |   size     = 6
----------------------------------------------
       ^-- start                               ^-- nextAvailPos

  0     1     2     3     4       5     6    7
----------------------------------------------   capacity = 8
| 15| -4  | 17 |        |      | 5    | 7 | 9 |   size     = 6
----------------------------------------------
 nextAvailPos --^            ^-- start
```

Circular Array (Iteration - Forward)

```
# Forward Iteration
def forwardIteration(cir, start, size):
    k = start
    for i in range(size):
        print(cir[k])
        k = (k + 1) % len(cir)
```

Circular Array (Iteration - Backward)

```
# Backward Iteration
def backwardIteration(cir, start, size):
  k = (start + size - 1) % len(cir)
  for i in range(size):
    print(cir[k])
    k = (k - 1) % len(cir)
```

Circular Array (Insertion)

Offset i index = (start + i) % len(cir)

Last index = (start + size - 1) % len(cir)

Next available index = (start + size) % len(cir)

Circular Array (Insertion)

```
Insert element 13 in position 3, or at index (5 + 3) % 8 = 0.

     0    1    2    3    4     5    6   7
   ----------------------------------------   capacity = 8
   | 15| -4 | 17 |     |     | 5    | 7 | 9 |   size     = 6
   ----------------------------------------
     ^                        ^-- start
     ^--- this is position 3 relative to front

And the resulting circular array after insertion is show below.

     0    1    2    3    4     5    6   7
   ----------------------------------------   capacity = 8
   |13 | 15 | -4 | 17  |     | 5    | 7 | 9 |   size     = 7
   ----------------------------------------
                              ^-- start
```

Circular Array (Insertion)

```
# Insert in Circular Array
def insert(cir_arr, start, size, elem, pos):
    if size == len(cir_arr):
        print("No empty Spaces")
    nShifts = size - pos
    fr = (start + size - 1) % len(cir_arr)
    to = (fr + 1) % len(cir_arr)
    for i in range(nShifts):
        cir_arr[to] = cir_arr[fr]
        to = fr
        fr = (fr - 1) % len(cir_arr)
    idx = (start + pos) % len(cir_arr)
    cir_arr[idx] = elem
    size += 1
    return cir_arr
```

Circular Array (Remove)

```
Remove element 15 in position 4, or at index (5 + 4) % 8 = 1.

   0    1    2    3    4     5    6   7
 ----------------------------------------   capacity = 8
 |13 | 15 | -4 | 17 |     | 5   | 7 | 9 |   size     = 7
 ----------------------------------------
                           ^-- start
       ^--- this is position 4 relative to front

And the resulting circular array after removal is show below.

   0    1    2    3    4     5    6   7
 ----------------------------------------   capacity = 8
 | 13| -4 | 17 |      |     | 5   | 7 | 9 |   size     = 6
 ----------------------------------------
                           ^-- start
```

Circular Array (Remove)

```python
# Remove value from circular array by left shift
def removeByLeftShift(cir_arr, start, size, pos):
  nShift = size - pos - 1
  idx = (start + pos) % len(cir_arr)
  removed = cir_arr[idx]
  to = idx
  fr = (to + 1) % len(cir_arr)
  for i in range(nShifts):
    cir_arr[to] = cir_arr[fr]
    to = fr
    fr = (fr + 1) % len(cir_arr)
  cir_arr[fr] = None
  size -= 1
  return removed
```

Queue (with Array)

```python
class ArrayQueue:
  def __init__(self):
    self.queue = [None] * 10 # Queue with size 10
    self.front = 0 # Initializing at index 0
    self.back = 0 # Initializing at index 0
    self.size = 0 # no elements
```

Queue (Enqueue)

```python
def enqueue(self, elem):
    if self.size == len(self.queue):
        print("Queue Overflow")
    else:
        self.queue[self.back] = elem
        self.back = (self.back + 1) % len(self.queue)
        self.size += 1
```

Queue (Dequeue)

```
def dequeue(self):
    if self.size == 0:
        print("Queue Underflow")
    else:
        dequeued_item = self.queue[self.front]
        self.queue[self.front] = None
        self.front = (self.front + 1) % len(self.queue)
        self.size -= 1
        return dequeued_item
```

Queue (Peek)

```
def peek(self):
    if self.size == 0:
        print("Queue Underflow")
    else:
        return self.queue[self.front]
```

Queue Simulation

enqueue a,

enqueue b,

dequeue,

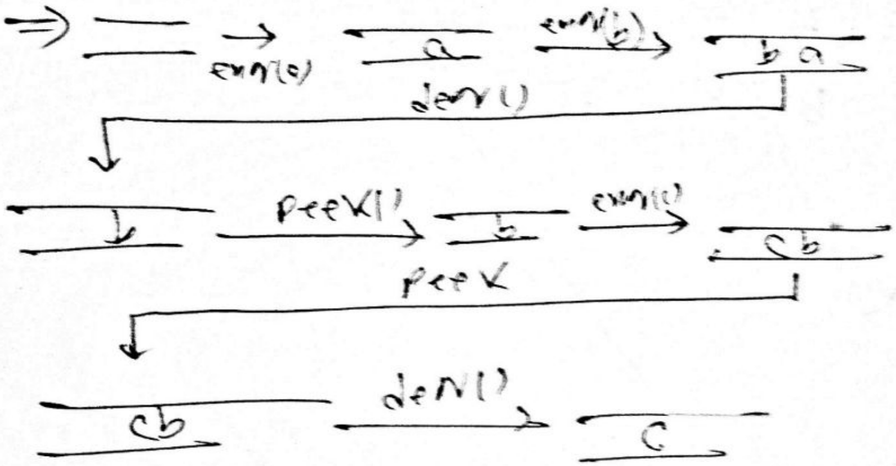
peek,

enqueue c,

peek,

dequeue

Queue Simulation



Dequeued : a, b

Peek: b, b

Queue Simulation

enqueue a,
enqueue b,
dequeue,
peek,
enqueue c,
peek,
dequeue

len(array) = 4

Start idx = 3

Queue Simulation

	0	1	2	3
	N	N	N	N
enq(a)	N	N	N	a
enq(b)	b	N	N	a
deq()	b	N	N	N
peek()	b	N	N	N
enq(c)	b	c	N	N
peek()	b	c	N	N
deq()	N	c	N	N

front	rear
~~3~~	~~3~~
~~0~~	~~0~~
1	~~1~~
	2

Dequeued: a, b

peek: b, b

After all operation,
front = 1
rear = 2